

# Predictive Maintenance and Anomaly Detection in 6G-Integrated Smart Manufacturing Systems

Detailed guide and project requirements for the Predictive Maintenance and Anomaly Detection in 6G-Integrated Smart Manufacturing Systems analysis.

## Background and Context

In intelligent factories:

- Machine failures are rare but extremely costly
- Waiting for efficiency to degrade is often too late
- Subtle sensor deviations usually appear before breakdowns

With 6G-enabled real-time data, manufacturers can shift from: Reactive maintenance → Predictive maintenance

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## Problem Statement

### ❖ **Manufacturing organizations face:**

- Unexpected machine breakdowns
- High maintenance and downtime costs
- Difficulty identifying early warning signals in noisy sensor data

### ❖ **Traditional rule-based thresholds fail because:**

- Normal behavior varies by machine and mode
- Early anomalies are subtle and multidimensional

## Dataset Fields Description:

| Column Name                   | Description                                                                |
|-------------------------------|----------------------------------------------------------------------------|
| Date                          | Calendar date on which the machine data was recorded.                      |
| Time                          | Timestamp indicating the exact time of data capture.                       |
| Machine_ID                    | Unique identifier assigned to each industrial machine.                     |
| Operation_Mode                | Current operating mode of the machine (e.g., normal, high-load).           |
| Temperature_C                 | Operating temperature of the machine in degrees Celsius.                   |
| Vibration_Hz                  | Machine vibration frequency measured in Hertz                              |
| Power_Consumption_kW          | Electrical power consumed by the machine in kilowatts.                     |
| Network_Latency_ms            | Communication latency experienced over the 6G network in milliseconds.     |
| Packet_Loss_%                 | Percentage of data packets lost during machine communication.              |
| Quality_Control_Defect_Rate_% | Percentage of defective units identified during quality checks.            |
| Production_Speed_units_per_hr | Number of units produced per hour by the machine.                          |
| Predictive_Maintenance_Score  | AI-derived score indicating the machine’s maintenance readiness.           |
| Error_Rate_%                  | Percentage of operational errors occurring during production.              |
| Efficiency_Status             | Target label classifying manufacturing efficiency as High, Medium, or Low. |

## Data Science Methodology (Step-by-Step)

- ❖ **Baseline Behavior Modeling**
  - Learn “normal” machine behavior per Machine\_ID
  - Capture normal ranges of temperature, vibration, power, and error patterns
- ❖ **Feature Engineering**
  - Sensor deviation from rolling baseline

- Vibration-to-power instability ratio
- Error escalation trend indicators
- Maintenance score decay patterns

#### ❖ **Anomaly Detection Modeling**

- Algorithms used:

Isolation Forest

Autoencoders (optional deep learning variant)

- These models identify:

Rare combinations of sensor & error behavior

Deviations unseen in normal operation

#### ❖ **Anomaly Scoring**

- Each record receives an anomaly score
- Higher score = more abnormal behavior

#### ❖ **Predictive Maintenance Risk Classification**

Anomaly scores are mapped to:

- Low Risk (normal behavior)
- Medium Risk (early warning)
- High Risk (urgent maintenance required)

#### ❖ **Temporal Risk Escalation Analysis**

Anomaly scores are mapped to:

- Track anomaly score trends over time
- Identify machines with rapidly increasing risk
- Detect cascading failure patterns

## Key Performance Indicators (KPIs)

| KPI                       | Description                              |
|---------------------------|------------------------------------------|
| Anomaly Score             | Degree of deviation from normal behavior |
| Maintenance Risk Level    | Low / Medium / High                      |
| High-Risk Machine List    | Priority maintenance targets             |
| Early Warning Lead Time   | Time before potential failure            |
| Downtime Prevention Index | Avoided breakdown estimate               |

## Streamlit Web Application Requirements

### Dashboard Modules

- ❖ **Predictive Maintenance Overview**
  - Risk distribution across machines
  - Count of high-risk assets
- ❖ **Machine Anomaly Dashboard**
  - Anomaly score trends per machine
  - Sensor deviation visualization
- ❖ **Maintenance Alert Panel**
  - High-risk alerts
  - Recommended inspection priority
- ❖ **Historical Risk Analysis**
  - Risk escalation timelines and Post-maintenance behavior comparison

## User Capabilities

- Machine selector
- Risk threshold slider
- Time window selector
- Operation mode filter

## **Deliverables and Submission**

- Research paper (EDA, insights, recommendations)
- Streamlit dashboard (live analytics)
- Executive summary for government stakeholders

## **Conclusion**

This project introduces preventive intelligence into 6G-enabled smart manufacturing. Unlike which focuses on classifying current efficiency, this project predicts future machine health risks by detecting abnormal behavior patterns long before efficiency declines or failures occur. This makes a maintenance-centric, risk-prediction system, fundamentally different in purpose, modeling approach, and business value

[Access Dataset](#)